

Validation of Large Ionospheric Gradient Observed in India on 15, February 2015

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Introduction

An extreme ionospheric spatial gradient of 631 mm/km was observed from HYDE and HYDR viewing PRN 07 in India around 16 UT on 15 February 2015 [1]. This large gradient observation is needed to be validated to real anomaly as opposed to the artifacts of erroneous measurements because it would have significant effects on the performances and the availability of GBAS. This paper revisits the 15 February 2015 ionospheric event in India and also verifies the ionospheric gradient observation.

Extreme Gradient Event

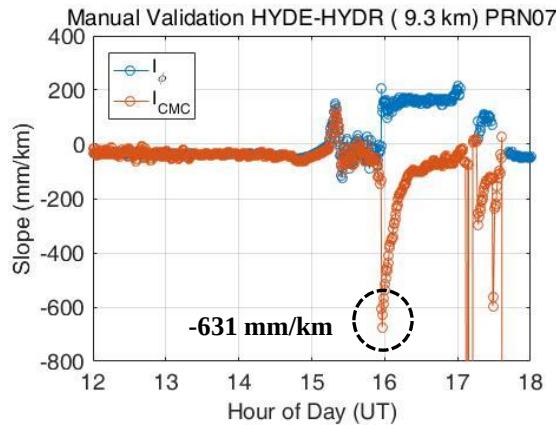


Figure 1. Extreme gradient event (-631 mm/km) observed from HYDE and HYDR. The red and blue cycles indicate the dual-frequency estimate of ionospheric gradient, and the single-frequency estimate of ionospheric gradient respectively

As shown in Figure 1, the single frequency estimate of ionospheric gradients reaches up to 631 mm/km around 16 UT, while the dual-frequency estimate remains at ~ 180 mm/km at that time. Since both estimates did not exhibit a similar gradient pattern, this extreme gradient observation in single frequency measurement should be validated.

Methodology

We used station HYDR only because the receiver at HYDE station was considered to have fault (the code measurement and the carrier-phase measurement from HYDE showed opposite pattern with each other). In addition, the baseline distances among nearby stations are too far apart compared with the spatial scale of the observed anomaly. Thus, this paper uses the ‘time-step method’ which require a single station and a single receiver as a pair to compensate for the lack of ionospheric observability. The ‘time-step method’ estimates ionospheric gradient with identical satellite and receiver pair. The gradient is computed by dividing the differential delay by the distance between ionospheric pierce points (IPP) at each epoch [2]. The satellites with similar azimuth may experience a similar ionospheric region to PRN 07. Therefore, PRN 28 and PRN 30, which have a similar azimuth angle with PRN 07, were used to validate the large gradient observation.

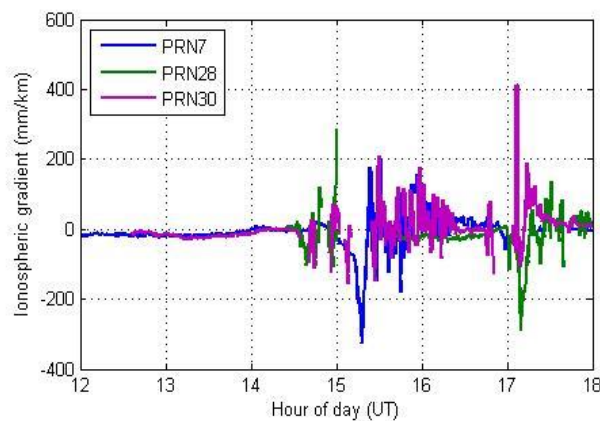


Figure 2. Ionospheric gradients of satellites (PRN 07, PRB 28, PRN 30) at HYDR

Results and Discussion

Figure 2 shows the ionospheric gradient estimates of PRN 07, PRN 28, and PRN 30 computed by the time-step method at HYDR station. All the satellites showed the severe ionospheric gradients larger than 200 mm/km. Especially for PRN 07, the maximum ionospheric gradient occurrence time was around 15 UT. In general, there is no significant difference between the occurrence times of the largest gradients computed by the time-step method and the station-pair method. However, the occurrence time with the time-step method was in discord with the occurrence time of the largest gradient computed by the station-pair method (i.e., ~ 16 UT). Actual ionospheric gradients might be less than 400 mm/km, because the ionospheric gradients computed by the time-step tend to be over-estimated than those computed by the station-pair method. The actual gradient value may be smaller than 400 mm/km. Therefore, the gradient observation of 631 mm/km could not be validated to be real. However, according to Figure 2, it is very likely that a few hundred mm/km (less than 400 mm/km) anomaly was occurred by a real event.

References

- [1] *Report of Ionospheric Studies Task Force Activities*, International Civil Aviation Organization, CNS SG/20-WP/10, Agenda Item 5.3, Bangkok, 11/07/16.
- [2] J. Lee, and S. Pullen, *Assessment of Ionosphere Spatial Decorrelation for Global Positioning System-Based Aircraft Landing Systems*, *Journal of Aircraft*, Vol. 44, No. 5, 2007, pp. 1662-1669.